Comparing hypothesis tests

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Hypothesis testing so far

General idea:

1. Have a parameter of interest, $\boldsymbol{\theta}$, and two competing hypotheses/models:

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \qquad H_A : \boldsymbol{\theta} \in \Theta_1$$

2. Find a statistic $T(\boldsymbol{X})$ that we expect to be "big" under H_A and "small" under H_0

3. Reject H_0 when $T(\boldsymbol{X})$ is "big". Specific rejection threshold chosen so that $\sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha$. And ideally want $\beta(\theta)$ large for $\theta \in \Theta_1$.

That's it! And there are *many* possible tests (and test statistics). The hard task is:

- Finding test statistics that behave the way we want (known distribution under H_0)
- Which test should we choose? I.e. which test has good power?

Some general-purpose tests

Wald test: Test hypotheses about a parameter $\boldsymbol{\theta}$ using asymptotic normality of estimator $\widehat{\boldsymbol{\theta}}$

$\theta \in \mathbb{R}$: $\hat{\theta} \approx N(\theta, S_n^2)$ $\quad Z = \dfrac{\hat{\theta} - \theta_0}{S_n} \approx N(0,1)$ under H_0

$\theta \in \mathbb{R}^q$: $\hat{\theta} \approx N(\theta, \Sigma_n)$ $\quad W = (\hat{\theta} - \theta_0)^T \Sigma_n^{-1} (\hat{\theta} - \theta_0)$

$\approx \chi^2_q$ under H_0

Likelihood ratio test: Test hypotheses about a parameter $\boldsymbol{\theta}$ using likelihood ratio

- $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $H_A : \boldsymbol{\theta} = \boldsymbol{\theta}_1 \implies$ Reject when $\dfrac{L(\boldsymbol{\theta}_1|\boldsymbol{X})}{L(\boldsymbol{\theta}_0|\boldsymbol{X})} > k$
- If distribution has MLR in a statistic $T(\boldsymbol{X})$, UMP test of $H_0 : \theta \leq \theta_0$ vs. $H_A : \theta > \theta_0$ rejects when $T(\boldsymbol{X}) > t_0$
- General case: reject when $\dfrac{\sup\limits_{\boldsymbol{\theta} \in \Theta_0} L(\boldsymbol{\theta}|\boldsymbol{X})}{\sup\limits_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta}|\boldsymbol{X})} \leq k$

$-2 \log \Lambda \xrightarrow{d} \chi^2_{d - d_0}$

Wald and likelihood ratio tests in the wild

Wald tests:

- Could view t tests as a variant of Wald tests that accounts for estimating the population variance
- One-sample tests, two-sampled tests, paired tests, etc.
- Tests for means and proportions
- Tests for individual regression coefficients (useful for testing directionality)
- Tests for M-estimators (likelihood doesn't apply)

Likelihood ratio tests:

- Chi-squared tests (independence, etc.)
- ANOVA, linear regression F tests
- Comparing nested models for GLMs
- In practice usually rely on asymptotic result $-2 \log \Lambda \approx \chi^2_{d-d_0}$ under H_0

Comparing Wald and LRT statistics

Suppose we observe data $X_1, ..., X_n$ from some distribution with parameter $\boldsymbol{\theta} \in \mathbb{R}^q$, and we wish to test

$$H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0 \qquad H_A : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$$

Consider three possible scenarios:

- H_0 is true: $\boldsymbol{\theta} = \theta_0$
- H_A is true, **fixed** alternative: $\boldsymbol{\theta} = \theta_0 + h$ $\quad (h \neq 0)$
- H_A is true, **local** alternative: $\boldsymbol{\theta} = \theta_0 + \frac{h}{\sqrt{n}}$ $\quad (h \neq 0)$

- $\beta(\theta_0) \approx \alpha$ (type I error) (for any reasonable test)
- If $\theta = \theta_0 + h$, expect $\beta(\theta) \to 1$ as $n \to \infty$ (for any reasonable test)
- If $\theta_n = \theta_0 + \frac{h}{\sqrt{n}}$, $\beta(\theta_n) \to ? \in (\alpha, 1)$ as $n \to \infty$

use <u>local</u> limiting power to compare tests

Comparing Wald and LRT statistics

Under H_0, or for a local alternative $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \frac{\boldsymbol{h}}{\sqrt{n}}$, Wald and LRT are asymptotically equivalent as $n \to \infty$ (under certain regularity conditions).

<u>Intuition</u>: Consider $\theta \in \mathbb{R}$. Previously, if $\hat{\theta} \approx \theta_0$ (either $\theta = \theta_0$ or $\theta = \theta_0 + \frac{h}{\sqrt{n}}$), then

$$\underbrace{2(\ell(\hat{\theta}) - \ell(\theta_0))}_{\text{LRT statistic}} \approx \underbrace{-\frac{1}{n}\ell''(\hat{\theta})}_{\substack{\mathcal{I}_1(\theta) \\ \approx \mathcal{I}_1(\theta_0)}} (\sqrt{n}(\hat{\theta} - \theta_0))^2$$

$$\approx \underbrace{\left(\frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\mathcal{I}_1^{-\frac{1}{2}}(\theta_0)}\right)^2}_{\text{Wald test statistic}}$$

Power under the alternative

Recall asymptotic normality of the MLE: $\widehat{\boldsymbol{\theta}} \approx N(\boldsymbol{\theta}, \mathcal{I}_n^{-1}(\boldsymbol{\theta}))$

Suppose we test $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $H_A : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$.

$$W = (\hat{\theta} - \theta_0)^T \mathcal{I}_n(\hat{\theta})(\hat{\theta} - \theta_0)$$

Under H_0, $W \approx \chi^2_q$ what happens under H_A??

<u>Def</u>: if $Z \sim N(\mu, I)$, $Z \in \mathbb{R}^q$, then

$$Z^T Z \sim \chi^2_q(\lambda) \qquad \lambda = \mu^T \mu$$

(non-central chi square with noncentrality parameter λ)

$$\hat{\theta} \approx N(\theta, \mathcal{I}_n^{-1}(\theta)) \Rightarrow \hat{\theta} - \theta_0 \approx N(\theta - \theta_0, \mathcal{I}_n^{-1}(\theta))$$

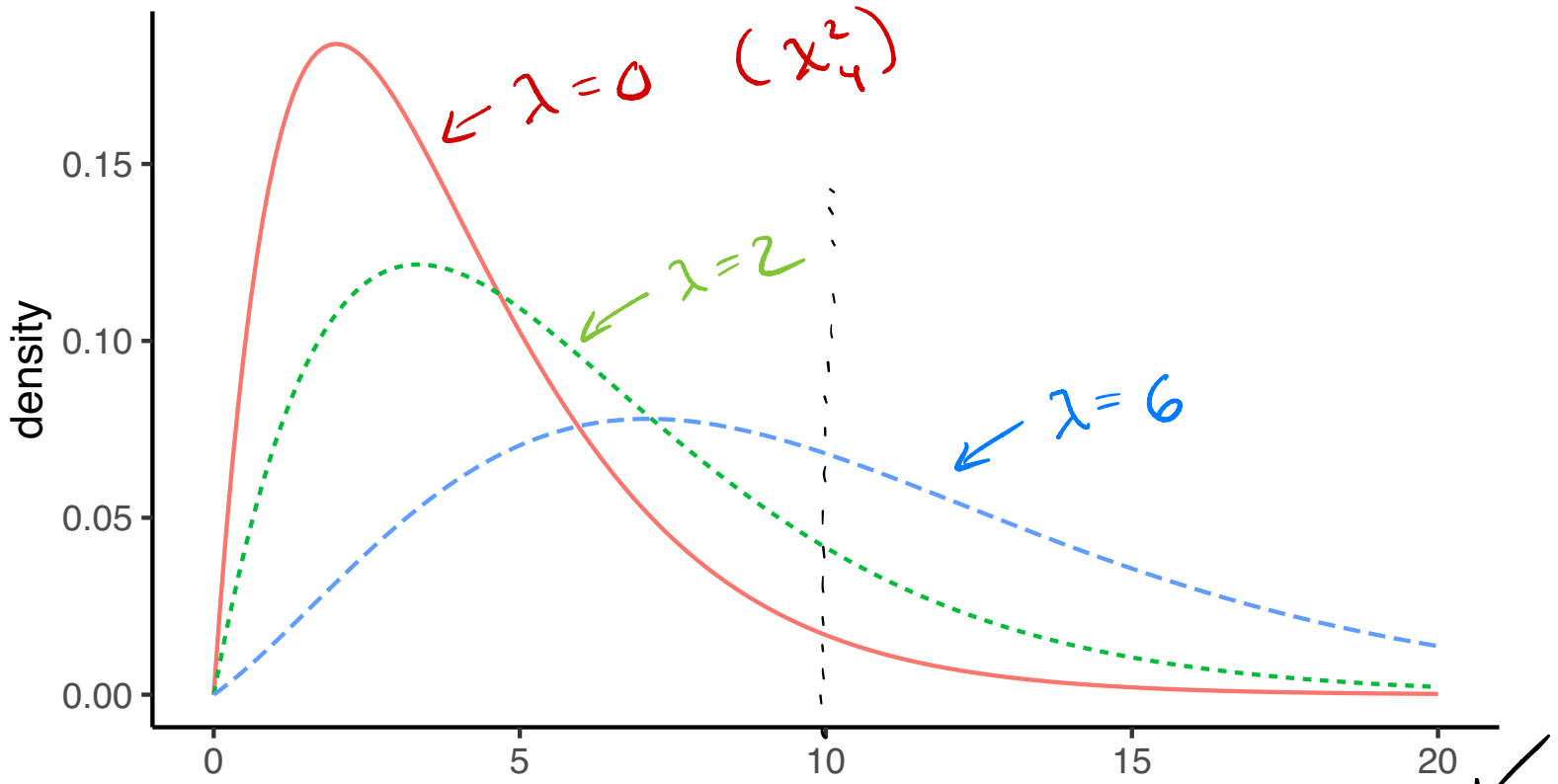
$$\Rightarrow \mathcal{I}_n^{\frac{1}{2}}(\hat{\theta})(\hat{\theta} - \theta_0) \approx N(\mathcal{I}_n^{\frac{1}{2}}(\theta)(\theta - \theta_0), I)$$

$$W = (\hat{\theta} - \theta_0)^T \mathcal{I}_n(\hat{\theta})(\hat{\theta} - \theta_0) \approx \chi^2_q(\lambda)$$

$$\lambda = (\theta - \theta_0)^T \mathcal{I}_n(\theta)(\theta - \theta_0)$$

$\chi^2_q(\lambda)$

$\chi^2_4(\lambda)$



upper α quantile of χ^2_q

$$\beta(\theta) = P_\theta(W > \text{cutoff}) = P_\theta(W > \chi^2_{q,\alpha})$$

$$= P_\theta(\chi^2_q(\lambda) > \chi^2_{q,\alpha})$$

As $\lambda \to \infty$, $\beta(\theta) \to 1$ (for any fixed n)

$$W = (\hat{\theta} - \theta_0)^T \mathcal{I}_n(\hat{\theta})(\hat{\theta} - \theta_0) \approx \chi^2_q(\lambda)$$

$$\lambda = (\theta - \theta_0)^T \mathcal{I}_n(\theta)(\theta - \theta_0)$$

If H_0 is true $(\theta = \theta_0)$: $\lambda = 0 \implies W \approx \chi^2_q(0) \equiv \chi^2_q$ ✓

Fixed alternative $(\theta = \theta_0 + h)$

$$\lambda = h^T \mathcal{I}_n(\theta) h$$

$$= \underset{\to \infty}{n} \underbrace{h^T \mathcal{I}_1(\theta) h}_{\text{fixed}}$$

$\mathcal{I}_n = n\mathcal{I}_1$ (iid data)

$\implies \lambda \to \infty$ as $n \to \infty$

so $\beta(\theta) \to 1$ as $n \to \infty$

Local alternative $\left(\theta = \theta_0 + \frac{h}{\sqrt{n}}\right)$

$$\lambda = \left(\frac{h}{\sqrt{n}}\right)^T \mathcal{I}_n(\theta)\left(\frac{h}{\sqrt{n}}\right)$$

$$= \frac{1}{n} \cdot n h^T \mathcal{I}_1(\theta) h$$

$$= h^T \mathcal{I}_1(\theta) h$$

$$\approx h^T \mathcal{I}_1(\theta_0) h$$

as $n \to \infty$, $\lambda \to h^T \mathcal{I}_1(\theta_0) h > 0$ $\quad \lambda \not\to \infty$

so $\beta(\theta) \to$ some stable value in $(\alpha, 1)$

Comparing other hypothesis tests

Suppose we have $X_1, ..., X_n$ iid from a symmetric distribution with mean and median θ. We wish to test

$$H_0 : \theta = 0 \qquad H_A : \theta > 0$$

Two asymptotic results:

Var(Xi)

sample median

$$\sqrt{n}(\overline{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2) \qquad \sqrt{n}(M_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{4f^2(\theta)}\right)$$

$\Rightarrow \frac{\sqrt{n}\,\bar{X}}{\sigma} \approx N(0,1)$ under H_0

reject when $\frac{\sqrt{n}\,\bar{X}}{\sigma} > z_\alpha$

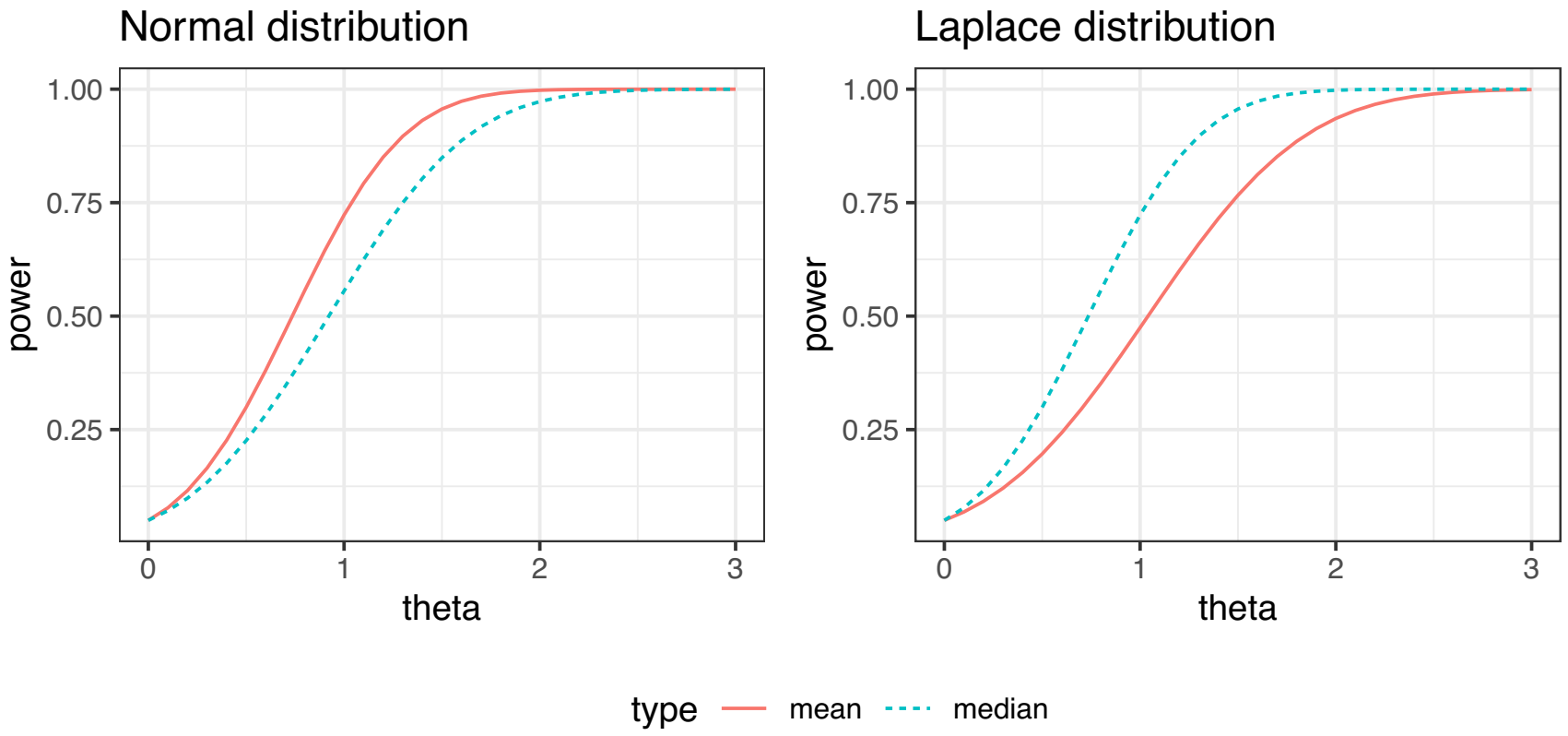
$\Rightarrow \sqrt{n}\, M_n \cdot 2f(M_n) \approx N(0,1)$ under H_0

reject when $\sqrt{n}\, M_n\, 2f(M_n) > z_\alpha$

Comparing other hypothesis tests

$$\sqrt{n}(\overline{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2) \qquad \sqrt{n}(M_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{4f^2(\theta)}\right)$$

$n = 20, \quad \alpha = 0.05$



Local asymptotic power

$$H_0 : \theta = 0 \qquad H_A : \theta > 0$$

Using the mean: $\beta_{mean}(\theta) \approx \Phi\left(\frac{\theta\sqrt{n}}{\sigma} - z_\alpha\right)$

Using the median: $\beta_{med}(\theta) \approx \Phi\left(2\theta f_\theta(\theta)\sqrt{n} - z_\alpha\right)$